FLATNESS OF A MODULE ON A RINGED SITE IS EQUIVALENT TO THE VANISHING OF TOR-1 AGAINST EVERY MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both directions: the forward implication is a one-line consequence of exactness, and the converse is carried out in full through the long exact sequence of Tor.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. The conclusion is the same and the ringed-space statement is the special case of this arbitrary-ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Flat module on a ringed site. Same chapter, the definition of a flat O-module as one for which tensorisation by F is exact. Supplies both the hypothesis of the forward implication and the conclusion of the converse.
- Tor functors of modules over a ringed site and their long exact sequence. Same chapter, the definition of Tor_i^O(F, -) as the derived functors of tensorisation, with the associated long exact sequence. Supplies Tor_1^O(F, Q) in the counted proof and the exactness sequence that identifies the kernel of the induced map.
- Right exactness of the tensor product. Same chapter, right exactness of F tensor_O - as a functor of the second variable. Makes the kernel of F tensor G -> F tensor H the relevant quantity to bound in the proof of the converse.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$ be an $\mathcal{O}$-module. The following are equivalent*

(1) *$\mathcal{F}$ is a flat $\mathcal{O}$-module, and*
(2) *$\mathit{Tor}_1^{\mathcal{O}}(\mathcal{F}, \mathcal{G}) = 0$ for every $\mathcal{O}$-module $\mathcal{G}$.*

Proof. If $\mathcal{F}$ is flat, then $\mathcal{F} \otimes_{\mathcal{O}} -$ is an exact functor and the satellites vanish. Conversely assume (2) holds. Then if $\mathcal{G} \to \mathcal{H}$ is injective with cokernel $\mathcal{Q}$, the long exact sequence of Tor shows that the kernel of $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G} \to \mathcal{F} \otimes_{\mathcal{O}} \mathcal{H}$ is a quotient of $\mathrm{Tor}_1^{\mathcal{O}}(\mathcal{F}, \mathcal{Q})$ which is zero by assumption. Hence $\mathcal{F}$ is flat. □

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.